Date.

From (26), one obtains $\frac{z(s)/s^3}{y(s)/s^2} \to -\frac{\tau_0}{3}$ as $s \to 0$

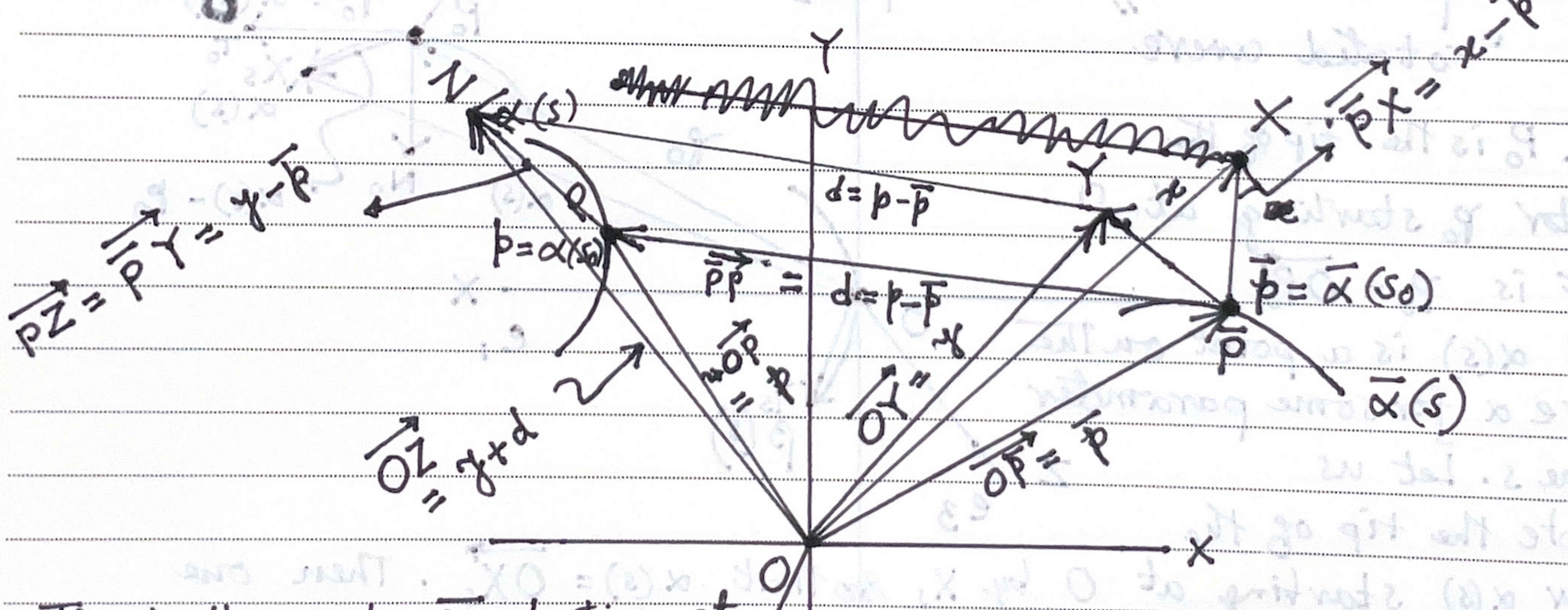
so that $c(s) \to 0$ as $s \to 0$; [From (29)]

Hence, the second direction $(0, 1, c(s))$ in (27) tends to e_2, while the first direction is always e_1. The limiting plane is therefore the XY plane, $z = 0$, or the osculating plane.

"End of lecture 4"

- Next lecture: global properties of plane curves.

Before embarking on global properties of plane curves, let us try to understand more explicitly what we did in the uniqueness proof of fundamental theorem and in the derivation of canonical form.



Y
V
α(s)
X
$\overrightarrow{\bar{p}X} = x - \bar{p}$
$\overrightarrow{pZ} = \overrightarrow{\bar{p}Y} = y - \bar{p}$
ρ
Y
x
$d = p - \bar{p}$
$p = \alpha(s_0)$
$\overrightarrow{\bar{p}p} = d = p - \bar{p}$
$\bar{p} = \bar{\alpha}(s_0)$
y
$\overrightarrow{OP} = p$
$\overrightarrow{OY} = y$
$\bar{p}$
$\bar{\alpha}(s)$
$\overrightarrow{OZ} = y + d$
$\overrightarrow{O\bar{P}} = \bar{p}$
O
X
Z

1. Tip of the vector $\bar{p}$ starting at O is $\bar{P}$. The geometric vector is denoted by $\overrightarrow{O\bar{P}} = \bar{p}$. The tip of the vector x starting at O is X. Here, $\overrightarrow{OX} = x$. $\overrightarrow{\bar{P}X} = x - \bar{p}$ Under rotation about $\bar{P}$, $\overrightarrow{\bar{P}X}$ is rotated to $\overrightarrow{\bar{P}Y} = \rho(x - \bar{p})$, whose tip is at Y. And $y = \overrightarrow{OY} = \overrightarrow{O\bar{P}} + \overrightarrow{\bar{P}Y} = \bar{p} + \rho(x - \bar{p})$ ....(1)

2. Translate by $d = p - \bar{p}$ which is both represented by $\overrightarrow{\bar{P}P}$ and its parallel translate $\overrightarrow{YZ}$. We have $\overrightarrow{PZ} = \overrightarrow{OZ} - \overrightarrow{OP} = (y + d) - (\bar{p} + d) = y - \bar{p}$. Also, from the triangle $O\bar{P}Y$, one has $\overrightarrow{\bar{P}Y} = y - \bar{p} = \overrightarrow{PZ}$.

Finally, Z is the image of X under the full rigid motion. Its position vector from O is

$$\vec{OZ} = y + d = \bar{p} + \rho(x - \bar{p}) + p - \bar{p}; \text{[using (1) and the fact that } d = p - \bar{p}]$$

$$= p + \rho(x - \bar{p}) = M(x)$$

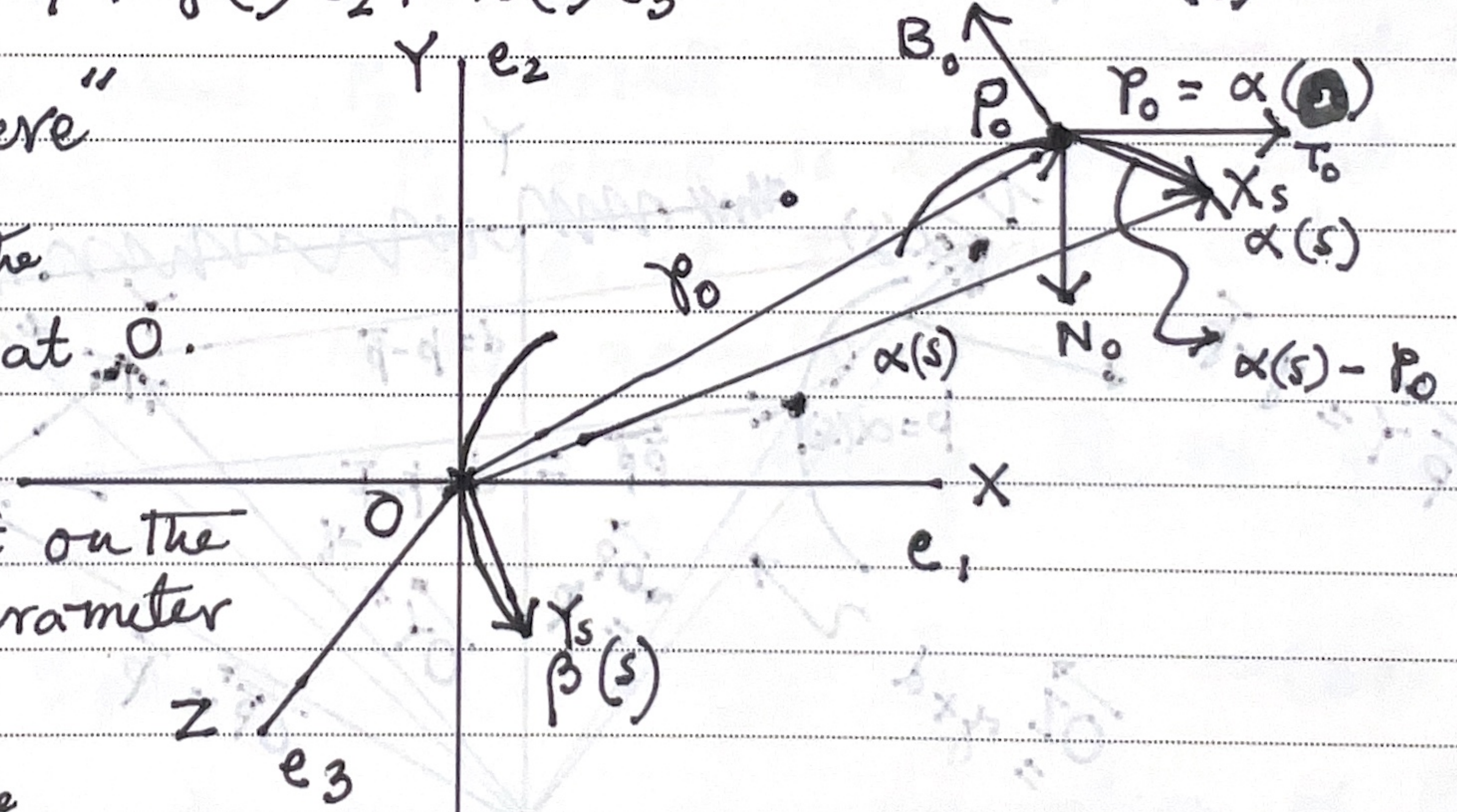
- Now come to the concrete geometric picture of the canonical form:

In lecture 4, we found

$\alpha(s) - p_0 = x(s)T_0 + y(s)N_0 + z(s)B_0$ ....(2)

and $\beta(s) = x(s)e_1 + y(s)e_2 + z(s)e_3$ ....(3)

"rotated curve"



P_0 is the tip of the vector p_0 starting at O. that is $p_0 = \vec{OP_0}$. And $\alpha(s)$ is a point on the curve α for some parameter value s. Let us denote the tip of the vector $\alpha(s)$ starting at O by X_s so that $\alpha(s) = \vec{OX_s}$. Then one reads off $\alpha(s) - p_0 = \vec{OX_s} - \vec{OP_0} = \vec{P_0X_s}$ ....(4)

In (2), we express this vector $\vec{P_0X_s}$ in the Frenet basis $\{T_0, N_0, B_0\}$ located at the point P_0. Now, for the same parameter value s $\beta(s)$ is the position vector with its tail at the origin O, i.e., $\beta(s) = \vec{OY_s}$, if we denote the ~~tail of~~ tip of the vector lying in $\beta(s)$ by Y_s. Therefore, $\alpha(s) - p_0$ is vector with its tail at P_0 while ~~$\beta(s)$ is a vector with its tail at O.~~

The coordinates of $\alpha(s) - P_0$ with respect to the Frenet basis $\{T_0, N_0, B_0\}$ at P_0 and those of $\beta(s)$ with respect to the standard basis $\{e_1, e_2, e_3\}$ at O exactly match given by (2) and (3). These coordinates are called the ~~canonical form of the curve $\alpha(s)$,~~ Frenet-adapted coordinates of $\alpha(s)$. Their Taylor expansions through cubic order, with remainders, give the local canonical form.

$\square$.